On the Guidance of Flow Matching: review on synthetic data

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Abstract

Flow matching has shown state-of-the-art performance in various generative tasks in image generation and scientific applicaiton. In many applicatons, it is desirable to generate specific parts of the target distribution which can be dynmaically controlled after training. Guidance methods have been proposed to steer a frozen flow matching model towards a defined objective. In this paper, we evaluate the guidance methods for flow matching on synthetic data and discuss the limiting assumptions of the methods.

1 Introduction

Deep generative models have achieved remarkable success across a wide range of domains, from high-fidelity image synthesis (Rombach et al., 2022) to computational biology (Klein et al., 2025) and drug discovery (Jumper et al., 2021). A shared principle behind these successes is the idea of learning a mapping from a simple, tractable distribution (e.g., standard Gaussian) to the complex target distribution of interest, enabling the model to generate realistic samples by transforming random noise. Diffusion models (Ho et al., 2020; Song et al., 2021; Rombach et al., 2022) realize this by learning to reverse a gradual noise-corruption process, and have set strong baselines across many benchmarks in computer vision. Flow matching (Lipman et al., 2023) approaches the same goal more directly. A time-dependent vector field is defined to continuously deform the prior distribution into the target data distribution, and a neural network is trained to approximate this field from data. At inference, new samples are generated by constructing a trajectory from a point in the source distribution to the target distribution through the learned vector field. This can be achieved by numerically integrating an ordinary differential equation (ODE). A critical design choice is the coupling, that is, how individual source samples from the prior are paired with target samples from the data distribution when constructing the training trajectories. Randomly coupling source and target points produces crossing trajectories that place conflicting demands on the vector field and make learning harder, whereas pairing via optimal transport yields shorter, straighter paths that are easier to approximate and require fewer ODE integration steps at inference (Tong et al., 2024). Entropic variants of this coupling have further extended flow matching to unpaired data settings such as single-cell genomics (Klein et al., 2024), broadening its practical applicability.

In many applications it is desirable to generate not just any sample from the target distribution, but samples that satisfy a user-specified criterion, for instance, images of a particular class. One natural approach is to incorporate this criterion at training time, for example by conditioning the model on a class label or other side information (Ho and Salimans, 2022); the resulting model can then generate samples from the desired subdistribution on demand. However, such training-time conditioning must be built into the model from the start, and obtaining samples under a new or different objective requires retraining.

A more flexible alternative is inference-time steering, commonly referred to as guidance. There, the generation process of an already trained model is modified at inference time to bias sampling toward

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the desired region of the target distribution, without any retraining (Dhariwal and Nichol, 2021; Feng et al., 2025). For diffusion models, this problem is well understood. Classifier guidance (Dhariwal and Nichol, 2021) and classifier-free guidance (Ho and Salimans, 2022) modify the score function at inference time to bias generation toward a desired condition. Transferring these ideas to flow matching is, however, non-trivial, because the deterministic trajectories and the absence of a stochastic noise schedule fundamentally change how an energy function can be incorporated into the velocity field. As a consequence, guidance for flow matching has remained largely underexplored compared to its diffusion counterpart. Feng et al. (2025) recently proposed the first general framework for guidance in flow matching, deriving a family of training-free and training-based techniques covering approximate and asymptotically exact methods. However, a systematic empirical comparison of these approaches and a characterization of their failure modes has not been carried out.

We summarize our contributions as follows:

- We provide a review of the guidance methods for flow matching.
- We experimentally test the guidance methods on synthetic data, visualize their behavior, and compare their performance in terms of different metrics.
- We experimentally examine the limitations of the guidance methods regarding different coupling strengths and desired subdistributions and their effect on the generated samples.

2 Background

2.1 Conditional Flow Matching (CFM)

Flow Matching (FM) is a generative method that learns a time-dependent velocity field $v_t(x)$ (with $t \in [0,1]$ and $x \in \mathbb{R}^d$) that transports a tractable source distribution p_0 (e.g. a standard Gaussian) into a complex target distribution p_1 on $\mathbb{R}^d$ (Lipman et al., 2023). The velocity field $v_t(x)$ is a map with $(t,x) \in [0,1] \times \mathbb{R}^d \to \mathbb{R}^d$ since the source and target distributions live in the same space. The velocity field defines a trajectory x_t starting from $x_0 \sim p_0$ through the ordinary differential equation (ODE)

$$\frac{d}{dt}x_t = v_t(x_t), \qquad x_t\big|_{t=0} = x_0. \tag{1}$$

Following this ODE over time from $t=0$ to $t=1$ continuously deforms the source into the target distribution. The distribution of x_t (denoted p_t) coincides with p_0 at $t=0$ and with p_1 at $t=1$. Thus $x_t\big|_{t=1} = x_1 \sim p_1$. A new sample is generated at inference time by drawing $x_0 \sim p_0$ and numerically integrating Eq. (1) forward to $t=1$.

Ideally v_t would be trained by regressing it onto the true marginal vector field $u_t(x)$ that generates the path p_t. This target has no closed form in general. Lipman et al. (2023) sidestep this by conditioning on a latent variable z such as an endpoint pair $z=(x_0,x_1)$ with $x_0 \sim p_0$ and $x_1 \sim p_1$. A conditional probability path $p_t(x|z)$ and a conditional velocity field $u_t(x|z)$ are then chosen in closed form. The marginal path is recovered as

$$p_t(x) = \int p_t(x|z)\, p(z)\, dz \tag{2}$$

and the marginal vector field is the conditional expectation of $u_t(x|z)$ over the posterior $p(z|x_t = x)$ which we can simplify with Bayes' rule to

$$u_t(x) = \int u_t(x|z)\, \frac{p_t(x|z)\, p(z)}{p_t(x)}\, dz. \tag{3}$$

Training v_t to match this marginal target directly amounts to minimizing

$$\mathcal{L}_{FM} = \mathbb{E}_{t\sim\mathcal{U}(0,1),\, x_t\sim p_t}\left[\|v_t(x_t) - u_t(x_t)\|^2\right]. \tag{4}$$

This expectation is intractable in practice. Evaluating $u_t(x_t)$ requires the integral in Eq. (3). Sampling $x_t \sim p_t$ also requires already knowing the path we are trying to learn. They show that regressing v_t onto the closed-form conditional target $u_t(x|z)$ yields the same gradient with respect to v_t as regressing onto the intractable marginal target in Eq. (4). The marginal objective can therefore be replaced by a tractable conditional counterpart. This yields the simulation-free Conditional Flow Matching (CFM) loss

$$\mathcal{L}_{CFM} = \mathbb{E}_{t\sim\mathcal{U}(0,1),\, z\sim p(z),\, x_t\sim p_t(\cdot|z)}\left[\|v_t(x_t) - u_t(x_t|z)\|^2\right], \tag{5}$$

where $u_t(x_t|z)$ is the conditional velocity field that transports x_0 to x_1 along the path defined by $z = (x_0, x_1)$. Furthermore, $v_t(x_t)$ is the neural network being trained to imitate the marginal vector field $u_t(x)$. Every variable in Eq. (5) is obtained without ever simulating the ODE in Eq. (1). The time t is drawn uniformly from $[0, 1]$. The pair $z = (x_0, x_1)$ is obtained by sampling $x_0 \sim p_0$ and $x_1 \sim p_1$ from the training data or jointly from a coupling as discussed in Section 2.2. Both x_t and $u_t(x_t|z)$ then follow directly from a closed-form conditional path given t and z which will be explained in the next paragraph. Finally $v_t(x_t)$ is simply a forward pass of the network being trained. Training therefore reduces to a plain regression problem.

There are many valid choices for the conditional path $p_t(x|z)$ and vector field $u_t(x|z)$ in Eq. (5). Throughout this paper we assume straight probability paths for all three coupling strategies discussed in Section 2.2. Each trajectory connects x_0 and x_1 along a straight line traversed at constant velocity

$$x_t = (1 - t)\, x_0 + t\, x_1, \qquad u_t(x_t|z) = x_1 - x_0 = \frac{x_1 - x_t}{1 - t}, \quad t \in [0, 1). \tag{6}$$

We work in the Euclidean space $\mathbb{R}^d$ where a straight line is the shortest connection between two points. Eq. (6) is therefore a reasonable default and is commonly referred to as the "optimal transport" (OT) path. The second equivalent form of u_t in Eq. (6) reappears directly in the velocity corrections used for guidance in Section 3. This choice is not forced by the framework itself. If the data instead lived on a curved manifold, a straight line in the ambient Euclidean space could leave it. Riemannian Flow Matching then replaces Eq. (6) with geodesics (the shortest paths on such spaces) (Chen and Lipman, 2024). Some applications instead call for a stochastic rather than a deterministic generative process. The conditional path can then be built from a Schrödinger bridge instead. This keeps an entropy-regularized transport structure closely related to the entropic coupling introduced in Section 2.2 while allowing genuine randomness along the trajectory (Tong et al., 2024).

2.2 Coupling

Equation (6) only fixes the shape of the path once the pair z is given. It says nothing about how x_0 and x_1 are paired up in the first place. That second choice is called the *coupling* π. A coupling π is any joint distribution over (x_0, x_1) with marginals p_0 and p_1 from which $z \sim \pi$ is drawn instead of the independent sampling $x_0 \sim p_0$ and $x_1 \sim p_1$ used above (Tong et al., 2024). Only the joint distribution π that produces the pair z changes between the three coupling strategies discussed below. The same linear path from Eq. (6) is used throughout. Figure 1 compares the resulting trajectories.

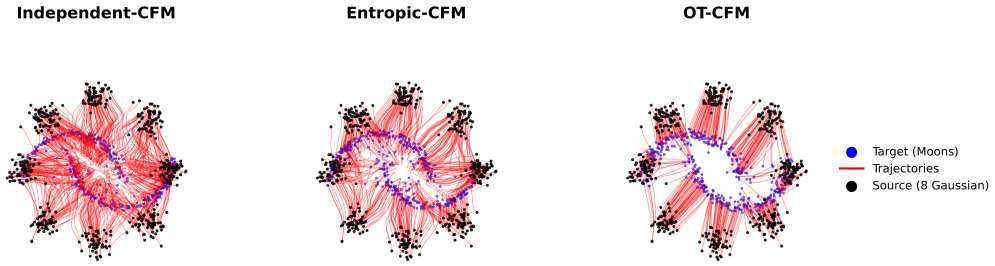


Figure 1: Comparison of three coupling strategies for flow matching: Independent-CFM, Entropic-CFM, and OT-CFM. Each panel shows sample trajectories (red) from the source distribution (8 Gaussians, black) to the target distribution (Moons, blue). OT-CFM produces shorter, straighter trajectories by minimizing transport cost, while Independent-CFM yields crossing paths.

2.2.1 Optimal Transport (OT) Coupling

The most natural choice follows directly from the motivation above. The OT coupling is the coupling π with marginal distributions p_0 and p_1 that minimizes the expected squared transport cost between x_0 and x_1

$$\pi^\star = \arg\min_{\pi} \; \mathbb{E}_{(x_0,x_1)\sim\pi} \left[\|x_1 - x_0\|^2 \right]. \tag{7}$$

We use the *minibatch OT* coupling of Tong et al. (2024): for each training batch of n source and n target points, Eq. (7) is solved exactly on just these $2n$ points with the network simplex algorithm, giving the cheapest one-to-one pairing within the batch. Sampling $z \sim \pi^\star$ then pairs each source point with the target point it can reach most cheaply, so trajectories stay short and rarely cross, as seen in the OT-CFM panel of Fig. 1. Such straight non-crossing paths are easier for v_t to approximate and require fewer ODE integration steps at inference (Tong et al., 2024).

2.2.2 Independent Coupling

The simplest and cheapest alternative avoids solving Eq. (7) altogether by taking the independent coupling $\pi(x_0, x_1) = p_0(x_0)\, p_1(x_1)$ (Tong et al., 2024). This pairs source and target points at random (Independent-CFM panel of Fig. 1). Because the pairing ignores distance, a source point can end up paired with a far-away unrelated target point. Trajectories can then cross one another and place conflicting demands on v_t at the same (t, x). Since v_t must assign a single velocity to every point (t, x), it cannot follow two crossing trajectories at once and instead ends up learning something close to their mean direction, distorting both. This makes the vector field harder to learn and typically requires more ODE integration steps at inference.

2.2.3 Entropic Coupling

Exact OT couplings are costly to recompute for every minibatch and can be unstable to estimate from a finite sample of an empirical distribution. The *entropic coupling* trades some of the straightness of Eq. (7) for robustness and speed by relaxing the transport problem with an added entropy term on π, weighted by a regularization strength $\lambda \in (0, \infty)$. This regularized problem is solved efficiently with the Sinkhorn algorithm rather than an exact assignment solver. As $\lambda \to 0$ the entropic coupling recovers the exact OT coupling, and as $\lambda \to \infty$ it approaches the independent coupling. For instance, we will use $\lambda \in \{0.1, 1.0, 10.0\}$ in our experiments. The intermediate trajectories of the Entropic-CFM panel of Fig. 1 reflect this. This construction extends naturally to unpaired settings such as single-cell genomics (Klein et al., 2024).

The coupling overall controls how straight and non-crossing the training trajectories are. Section 4.2 shows that this choice directly affects how well an already trained flow can be steered by inference-time guidance.

2.3 Class-conditional Flow Matching (CCFM)

The guidance methods introduced in Section 3 steer a flow matching model $v_t(x)$ at inference time, without the model ever seeing the desired subdistribution during training. As a training-time alternative, we consider Class-conditional Flow Matching (CCFM), which instead conditions the velocity field on a discrete label, analogous to conditional generation in diffusion models (Ho and Salimans, 2022) and to class-conditional flow-based generative models trained on labeled image data (Ma et al., 2024). Each target sample x_1 is assigned a label $y = c(x_1) \in \{1, \dots, K\}$ by a fixed partition c of the target distribution. Crucially, y requires no additional sampling step. It is a deterministic function of x_1, which is already part of the pair $z = (x_0, x_1)$ drawn for the ordinary CFM loss, so y is fixed as soon as z is, before x_t is drawn from the conditional path $p_t(\cdot|z)$. The conditional velocity field $v_t(x, y)$ is trained by minimizing

$$\mathcal{L}_{CCFM} = \mathbb{E}_{t\sim\mathcal{U}(0,1),\, z=(x_0,x_1)\sim p(z),\, x_t\sim p_t(\cdot|z)} \left[\| v_t(x_t, c(x_1)) - u_t(x_t|z) \|^2 \right], \tag{8}$$

which is identical to Eq. (5) except that the network additionally receives the label $c(x_1)$ as input. The coupling and path construction of Section 2.2 are unaffected, but the label must be carried through the reshuffling induced by couplings so that $c(x_1)$ stays attached to its source-target pair z.

At inference, a sample from a desired subdistribution is generated by drawing y from the label distribution of that subdistribution and integrating Eq. (1) with $v_t(\cdot, y)$ in place of $v_t(\cdot)$, i.e., no test-time modification of the vector field is required. This makes CCFM substantially less general than the guidance methods of Section 3. It requires the label partition c to be fixed in advance and a separate model to be retrained whenever the target subdistribution changes, and it can only reach subdistributions expressible as a mixture of the fixed classes. In exchange, because the conditioning is learned rather than approximated at inference, we use CCFM in Section 4 as an oracle baseline against which the training-free guidance methods are compared.

3 Guidance

3.1 Monte Carlo (MC) estimation

3.2 Simulated-based Monte Carlo estimation

3.3 Covariance Guidance (cov-A / cov-G)

3.4 Learned guidance (GM)

3.4.1 Discussion on guidance methods

The MC method needs only minimal assumptions and minimal tuning in terms of hyperparameter but still produces solid results. However, the MC method does not scale well to higher dimensions because the variance explodes.

The learned (GM) model performs best in terms of W2 distance and energy distance, but several models need to be trained (first disadvantage) and one has to do so for each new energy function (second disadvantage).

The disadvantage of the covariance methods are the additional assumptions that are made, which can be violated in practice.

How does the bias of the guidance methods look like? Maybe something like mode collapse in the Cov methods

3.4.2 Synthetic data experiments

We evaluate the guidance methods on synthetic data and compare the results in terms of W2 distance, energy distance, divergence trace ratio and off-manifold rate. The 2D synthetic data allows to have interpretable results and visualize the generated samples compared to other complex real-world tasks like image generation or single-cell data.

4 Results

4.1 Comparison of guidance methods

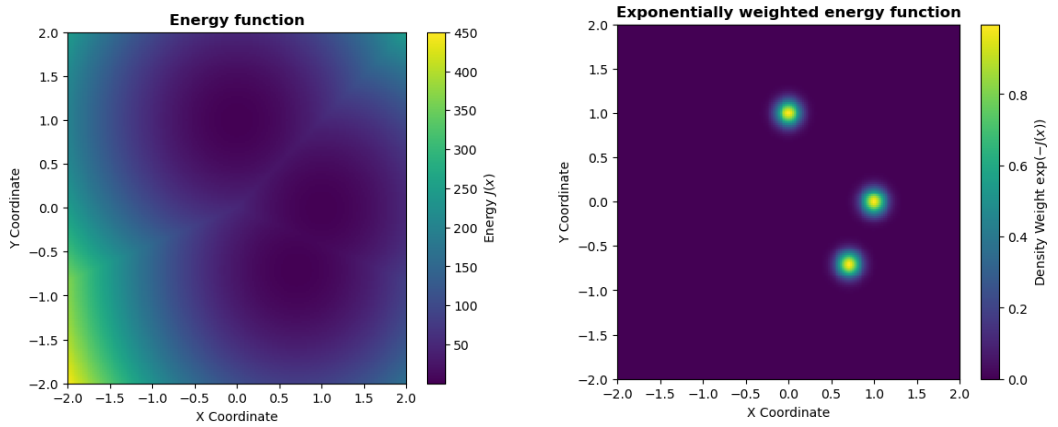
We define the energy function J to select three of the eight modes of the target distribution. For this, we define the energy function J as the sum of the squared distances to the three selected modes, scaled by a factor σ^2, which is the energy width.

$$J(x) = -\log \sum_{i \in 0,2,3} \exp\left(-\frac{|x - c_i|_2^2}{2\sigma^2}\right)$$

$$J(x) = \frac{|x - c_0|_2^2}{2\sigma^2} + \frac{|x - c_2|_2^2}{2\sigma^2} + \frac{|x - c_3|_2^2}{2\sigma^2}$$

Table 1: Comparison of methods across various metrics. The best method for each metric is **bolded**, and the second best is underlined.

Method	W2 ↓	Energy Dist ↓	Div Trace Ratio	Off-Manifold Rate ↓
Learned (GM)	**0.0802**	**0.0006**	0.9757	0.0659
Class-cond. CFM	<u>0.1632</u>	0.0033	<u>1.0069</u>	0.0786
MC	0.1698	<u>0.0030</u>	**0.9969**	0.0801
cov-G	0.2772	0.0296	1.2113	<u>0.0068</u>
sim-MC	0.3908	0.0739	0.7570	0.5098
cov-A	0.7905	0.2578	0.5410	**0.0008**
CFM	0.8852	0.3804	1.5355	0.6782



(a) Energy function $J(x)$.

(b) Exponentially weighted energy $e^{-J(x)}$.

Figure 2: Left: the energy function $J(x)$ selecting three of the eight Gaussian modes (low values indicate preferred regions). Right: the corresponding exponentially weighted density $e^{-J(x)}$, proportional to the guided target $p'(x) \propto p_1(x)\, e^{-J(x)}$, with the three target modes clearly localized.

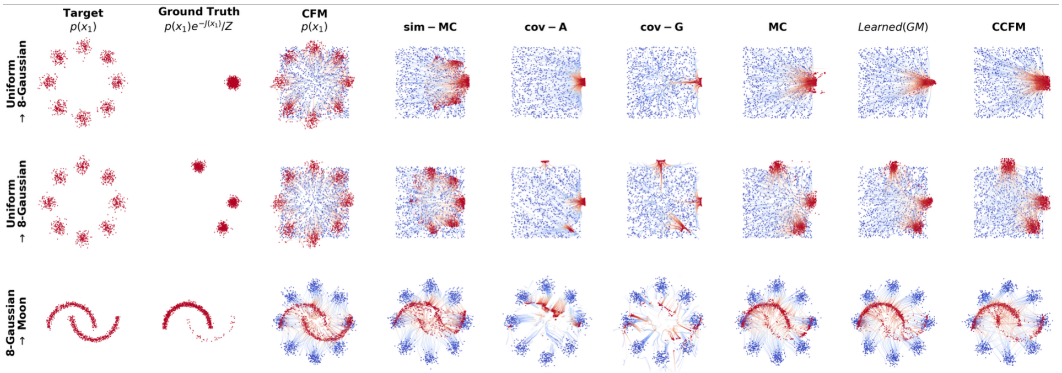


Figure 3: Qualitative comparison of guidance methods across three energy functions (rows). Columns show, from left to right: target distribution, ground truth $p'(x)$, and samples generated by each guidance method (CFM, sim-MC, cov-A, cov-G, MC, Learned (GM), CCFM). Red dots indicate samples in the guided target region; blue dots are the remaining samples.

4.2 Limitations of guidance

There are several limitations of guidance methods. Since the MC method is the best inference-steering method, we focus on the limitations of the MC method in this section. In many applications, the OT-coupling is used to learn the velocity field

4.3 Tuning hyperparameters

We want to discuss the effects of the hyperparameters on the guidance methods.

4.3.1 ODE solver

We increased the number of steps in the ODE solver from 100 to 200. This increased the performance of the guidance methods. Moreover, one could use a different ODE solver, e.g. Runge-Kutta 4th order, which might improve the generated samples further with the downside of increased computational cost. An early stopping criterion could be used to mitingate the problem of numerical instabilities at $t = 1$ and mode collapse in the MC guidance method.

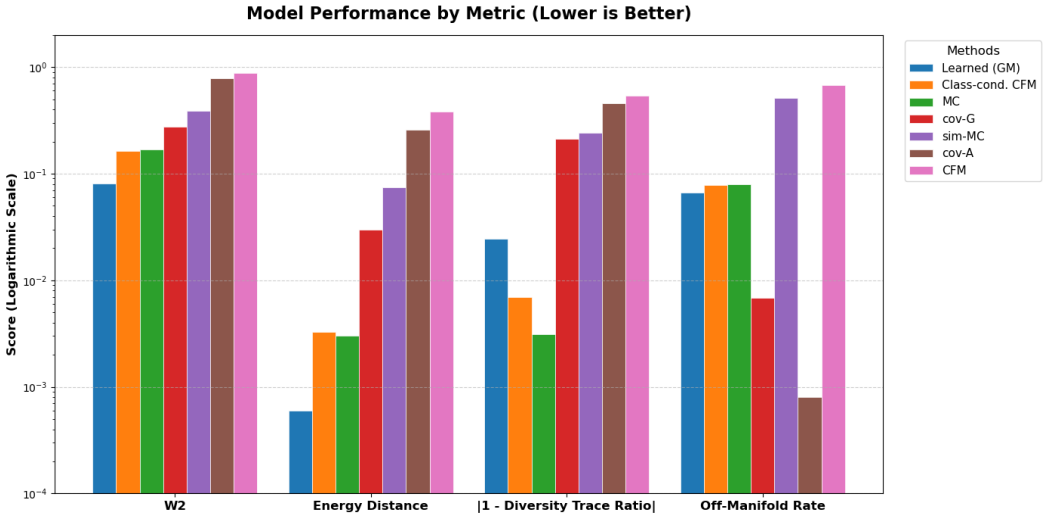
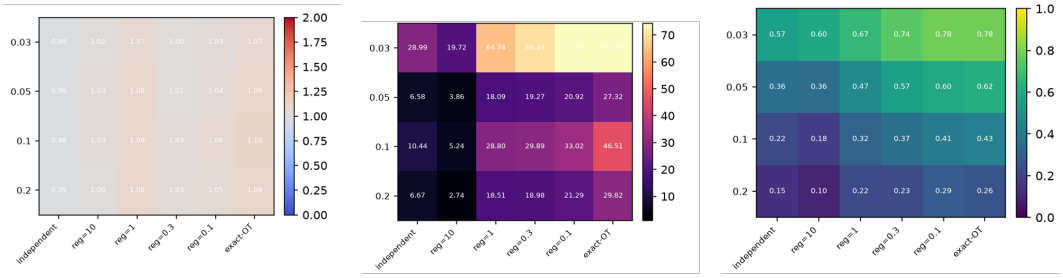


Figure 4: Quantitative comparison of all guidance methods on four metrics shown on a logarithmic scale. Lower is better for all metrics: W2 distance, energy distance, $|1 -$ diversity trace ratio$|$, and off-manifold rate. The learned (GM) model achieves the lowest energy distance; MC guidance offers a strong balance across all metrics without requiring additional training.



(a) Diversity trace ratio. (b) Energy distance. (c) Off-manifold rate.

Figure 5: Phase diagrams for MC guidance across coupling regularization strengths (columns, from independent coupling to exact-OT) and the stabilization parameter ϵ (rows, from 0.03 to 0.2). Stronger coupling (right columns) consistently increases the energy distance and off-manifold rate, indicating that the independence assumption of MC guidance breaks down under tight couplings.

4.3.2 MC Guidance

The parameter `mc_ep` (ϵ) appears in the denominator $(1 - t + \epsilon)$ of both the MC importance weights and the velocity correction $u = (x_1 - x_t)/(1 - t + \epsilon)$, acting as a numerical stabilizer as $t \to 1$. Without sufficient ϵ, back-projected source points fall outside the prior support, causing all importance weights to clamp to a uniform distribution and injecting large random forces that scatter particles off the manifold. We tried $\epsilon \in \{0.001, 0.005, 0.01, 0.05, 0.1\}$, with results shown in Table 2. Hence, we increased `mc_ep` from 0.001 to 0.05, which reduced the numerical instability and the outliers of the generated distribution significantly.

Table 2: Effect of the `mc_ep` parameter ϵ on sample quality.

ϵ	W_2 Distance ↓	Off-Manifold Rate ↓
0.001 (default)	0.4244	0.266
0.010	0.3053	0.042
0.050 (tuned)	**0.1091**	**0.002**
0.100	0.2032	0.004

4.3.3 Cov Guidance

5 Conclusion

Conclusion

$p'(x) = \frac{1}{Z} p_1(x) e^{-J(x)}$

Acknowledgments and Disclosure of Funding

Acknowledgement

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A Appendix

Appendix